YASWANT VADDI

+1 873-455-5577 | vaddiyaswant@gmail.com | yaswantvaddi | Google Scholar

Experimental nanophotonics researcher with a **PhD in Physics** and **6+ years of experience** in photonic device design, cleanroom nanofabrication, optical characterization, and electromagnetic simulation. Experienced in developing **plasmonic nanocavities, metasurfaces, nanoantennas, ENZ thin films, and multilayer photonic structures** through iterative simulation, fabrication, and experimental validation. Current research focuses on **strong light–matter coupling and quantum nanophotonic devices**. Skilled in **Lumerical FDTD/MODE, COMSOL, KLayout, Python, and LabVIEW**, with extensive hands-on experience in **nanophotonic device fabrication, optical experimentation, and simulation-guided device development**.

Technical Skills

- **Nanofabrication & Semiconductor Processing:** Electron-beam lithography, nanophotonic device fabrication, plasmonic nanoantennas and metasurfaces, multilayer thin-film stacks, electron-beam evaporation, thermal evaporation, RF/DC sputtering, metal lift-off, KLayout, process optimization, and iterative device fabrication.
- **Materials & Thin Films:** ITO and ENZ materials, Si, SiO_2, Au, dielectric/metal multilayers; thin-film deposition, spectroscopic ellipsometry, optical-constant and thickness characterization, and correlation of material properties with photonic-device performance.
- **Quantum & Nanophotonics:** Strong light–matter coupling, quantum-emitter–nanocavity interactions, plasmonic nanocavities, metasurfaces, nanoantennas, resonant photonic structures, cavity resonance engineering, mode coupling, field confinement and enhancement, nonlinear optics, and quantum sensing.
- **Photonic Design & Simulation:** Lumerical FDTD, Lumerical MODE, COMSOL Multiphysics, KLayout; electromagnetic modeling of nanophotonic and resonant structures, optical mode and waveguide analysis, near-field distributions, coupling analysis, optical loss identification, and simulation-guided device optimization.
- **Optical Characterization:** Free-space optical alignment, polarization optics, resonance spectroscopy, femtosecond pump-probe spectroscopy, wavelength-, angle-, polarization-, and power-dependent characterization, optical spectrum analyzers, photodetectors, power meters, SourceMeters, probe stations, and spectroscopic ellipsometry.
- **Automation & Programming:** Python (PyVISA, Pandas, NumPy), LabVIEW, MATLAB, Mathematica, C/C++, Git, NI DAQ systems; automated instrument control, data acquisition, analysis, visualization, and experimental workflow development.

Education

University of Ottawa, Ottawa, Ontario, Canada — Sep 2019 – Jul 2026
Doctor of Philosophy, Physics

Indian Institute of Science, Bangalore, India — Jun 2015 – Jun 2016
Master of Science, Physics

Indian Institute of Science, Bangalore, India — Aug 2011 – Jun 2015
Bachelor of Science Research, Physics

Professional & Research Experience

Postdoctoral Fellow — National Research Council Canada (NRC) — Aug 2026 – Present
Metrology Research Centre, Ottawa, ON

Project: Quantum Plasmonic Sensors for Chemical & Biological Sensing

- **Quantum nanophotonic device development:** Developing plasmonic nanoantenna and nanocavity platforms for **strong light–matter coupling with quantum emitters**, targeting quantum sensing and cavity-enhanced light–matter interactions.
- **Nanophotonic design and simulation:** Designing Au bow-tie nanoantennas and related plasmonic resonators using electromagnetic simulations, optimizing resonance wavelength, nanoscale gap size, near-field enhancement, mesh convergence, and fabrication tolerances.
- **Nanofabrication process development:** Developing electron-beam lithography and metal lift-off processes for nanoscale Au structures, including substrate selection, resist-stack optimization, patterning, metal deposition, and device characterization.

Ph.D. Researcher, Experimental Nanophotonics — University of Ottawa — Sep 2019 – Jul 2026
Supervisor: Dr. Robert W. Boyd

Dissertation: Enhanced Optical Nonlinearities in Epsilon-Near-Zero Nanomaterials

- **Photonic device design, simulation, and characterization:** Designed, simulated, fabricated, and characterized ENZ nanocavities, plasmonic metasurfaces, and multilayer photonic structures. Compared experimental measurements with electromagnetic simulations using **Lumerical FDTD, Lumerical MODE, and COMSOL** to validate device performance, analyze optical modes and coupling, identify loss mechanisms, and guide design improvements. Developed repeatable wavelength-, angle-, power-, and polarization-dependent measurements to characterize resonant photonic devices.

- **Optical experimentation, alignment, and automation:** Designed, built, aligned, and optimized complex free-space optical setups involving femtosecond laser sources, beam-shaping optics, polarization optics, delay stages, detectors, and spectrometers. Developed automated measurement frameworks using **Python/PyVISA, LabVIEW, Pandas, and NumPy** for instrument control, data acquisition, visualization, and analysis.
- **Strongly coupled ENZ plasmonic nanocavity for high-efficiency four-wave mixing:** Designed, simulated, fabricated, and characterized a 2-μm-thick ITO-SiO_2-Au Gires–Tournois plasmonic nanocavity, achieving **up to 45% four-wave-mixing conversion efficiency**. Used iterative cycles of simulation, fabrication, and characterization to optimize cavity resonance conditions, optical coupling, field confinement, and nonlinear conversion efficiency.
- **Ultrafast cavity-enhanced optical switching:** Demonstrated **25 dB modulation on a sub-picosecond timescale** using femtosecond pump-probe spectroscopy. Characterized temporal response, spectral dynamics, resonance behavior, and measurement repeatability of the switching process.
- **Dynamic coherent perfect absorption in nonlinear metasurfaces:** Designed and modeled ultrathin plasmonic resonators integrated with ENZ substrates using **COMSOL**. Investigated phase-controlled and pump-tunable optical response through mode interference, resonance engineering, and optical coupling analysis.
- **Parametric amplification in time-varying ENZ films:** Demonstrated efficient two-beam coupling and coherent energy transfer, with energy-transfer efficiencies up to **10 orders of magnitude higher than conventional Kerr media**. Validated the observed amplification through experimental measurements and numerical modeling.
- **Nanofabrication and process integration:** Fabricated Au nanoantennas, ENZ thin films, and multilayer cavity stacks using electron-beam evaporation, thermal evaporation, and RF/DC sputtering. Correlated material properties, fabrication parameters, electromagnetic simulations, and measured optical response to identify device limitations, refine fabrication targets, and guide subsequent device iterations.

Previous Research Experience — Indian Institute of Science 2014 – 2017

- **Flexible thin-film transistor fabrication and characterization:** Developed flexible metal–insulator–metal capacitors and optimized PVDF dielectric thickness, roughness, leakage characteristics, and mechanical stability for thin-film transistor architectures.
- **Self-healing interconnects and reconfigurable electronics:** Investigated self-assembly of conductive particles and nanotubes for automatic repair of open-circuit faults. Demonstrated rewritable electrical pathways with memory-like behavior.

Selected Publications

- **Yaswant Vaddi**[†], Theng-Loo Lim[†], Rasoul Alaee, Shivashankar Vangala, Jeremy Upham, Joshua R. Hendrickson, M. Zahirul Alam, and Robert W. Boyd. Unity-order four-wave mixing efficiencies in epsilon-near-zero nanocavities. Under Review in Nature Communications
- Theng-Loo Lim[†], **Yaswant Vaddi**[†], Rasoul Alaee, Shivashankar Vangala, Jeremy Upham, Joshua R. Hendrickson, M. Zahirul Alam, and Robert W. Boyd. Spectral reshaping in a dynamic epsilon-near-zero-based Gires–Tournois nanocavity. Under Review in Advanced Materials
- **Yaswant Vaddi**, Theng-Loo Lim, Yiyu Zhou, Mohammad Karimi, Shivashankar Vangala, Jeremy Upham, Joshua R. Hendrickson, Robert W. Boyd, and M. Zahirul Alam. Parametric amplification of optical pulses through synthetic motion in a time-varying medium. Nano Letters (2026)
- Mohammad Karimi, Kashif Masud Awan, **Yaswant Vaddi**, Rasoul Alaee, Jeremy Upham, M. Zahirul Alam, and Robert W. Boyd. Interactions of fundamental Mie modes with thin epsilon-near-zero substrates. Nano Letters **23**(24), 11555–11561 (2023)
- Theng-Loo Lim, **Yaswant Vaddi**, M. Saad Bin Alam, Lin Cheng, Rasoul Alaee, Jeremy Upham, Mikko J. Huttunen, Ksenia Dolgaleva, Orad Reshef, and Robert W. Boyd. Fourier-engineered plasmonic lattice resonances. ACS Nano **16**(4), 5696–5703 (2022)
- Rasoul Alaee, **Yaswant Vaddi**, and Robert W. Boyd. Dynamic coherent perfect absorption in nonlinear metasurfaces. Optics Letters **45**(23), 6414–6417 (2020)
- **Yaswant Vaddi**, A. Kumar, and Sanjiv Sambandan. Self-healing of open circuit faults: with active reconfigurability and mimicry of synaptic plasticity. Applied Physics Letters **109**(2), 024101 (2016)
- S. R. Udatha, A. Ruhela, G. Saravanavel, **Yaswant Vaddi**, J. Singh, and Sanjiv Sambandan. Design optimization of thin-film transistors based on a metal-substrate-semiconductor architecture for high DC voltage sensing. IEEE Transactions on Electron Devices **63**(4), 1696–1703 (2016)
- A. Nair, K. Raghunandan, **Yaswant Vaddi**, S. S. Pillai, and Sanjiv Sambandan. Maze-solving automatons for self-healing of open interconnects: modular add-on for circuit boards. Applied Physics Letters **106**(12), 123103 (2015)

Selected Honors

- Invited talk: "Dynamic Coherent Perfect Absorption using Epsilon-Near-Zero-Based Nonlinear Plasmonic Metasurfaces" at **Photonics North Conference, Quebec, Canada** (2020).
- Ranked **122 (All India)** in CSIR-UGC NET (Physics) (2016); recipient of **INSPIRE SHE Fellowship** from the Department of Science and Technology (DST), India (2011–2016).